

# Nicholas Zolman

PHD CANDIDATE · MECHANICAL ENGINEERING

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## Research Statement

*An interdisciplinary researcher working at the intersection of scientific machine learning, dynamics, control theory, and interpretable low-order modeling with a commitment to building community in open, inclusive, and collaborative environments.*

## Education

### University of Washington

PHD, MECHANICAL ENGINEERING

- Advisor: Professor Steven L. Brunton
- Topic: Parsimonious Priors for Planning, Projections, and Physics
- MS awarded 2024

Seattle, WA

2022–2027 (expected)

### California Institute of Technology

BS, MATHEMATICS

- Graduated with Honors, emphasis on physics.

Pasadena, CA

2012–2016

## Professional Experience

2022– **Graduate Research Assistant**, University of Washington

2018–2025 **Data Scientist**, The Aerospace Corporation

- Proposed and awarded research funding for various projects: e.g. physics-informed ML, digital twins development, anomaly detection, and DRL for satellite operations.
- Led interdisciplinary teams of engineers and subject matter experts.
- Presented findings to leadership and government stakeholders.

2017–2018 **Mathematics Graduate Teaching Assistant**, UCLA

2017 **Data Science Intern**, The Aerospace Corporation

2016–2017 **Residential Counselor**, Gatton Academy of Mathematics and Science

- Lived in a residential hall on Western Kentucky University's campus with motivated STEM high school students while they took WKU classes.
- Served as a form of emotional support for students as they endured this challenging environment, while also acting as an academic support by tutoring physics, computer science, and math classes.

2016 **Undergraduate Research Assistant**, Economics, California Institute of Technology

- Worked with Professor Antonio Rangel on Bayesian methods for identifying optimal student curricula.

2015 **Summer Undergraduate Research Fellow**, Mathematics, California Institute of Technology

- Worked with Professor Matilde Marcolli on geometric methods for supersymmetry algebras.

2014 **Summer Undergraduate Research Fellow**, Astrophysics, California Institute of Technology

- Worked with Professor Philip Hopkins on analyzing galaxy mass-metallicity relation from large suite of galaxy simulations.

2013 **Summer Undergraduate Research Assistant**, Physics, Western Kentucky University

- Worked with Professor Keith Andrew on modified gravity theories.

## Publications

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### PUBLISHED

- Lagemann, C., Mokbel, S., Gondrum, M., Rüttgers, M., Wang, Y., Suárez, P., Paehler, L., Bezgin, D. A., Buhendwa, A. B., Callahan, J. L., Ahnert, S., **Zolman, N.**, Shao, X., Loiseau, J.-C., Adams, N. A., Meinke, M., Schröder, W., Lagemann, K., Lagemann, E., ... Brunton, S. L. (2026). The HydroGym reinforcement learning platform for fluid dynamics. *Nature*. <https://doi.org/10.1038/s41586-026-10917-6>
- Brunton, S. L., **Zolman, N.**, Kutz, J. N., & Fasel, U. (2025). Machine learning for sparse nonlinear modeling and control. *Annual Review of Control, Robotics, and Autonomous Systems*, 8(1), 127–152.
- Otto, S. E., **Zolman, N.**, Kutz, J. N., & Brunton, S. L. (2025). A unified framework to enforce, discover, and promote symmetry in machine learning. *Journal of Machine Learning Research*, 26(248), 1–83.
- Zolman, N.**, Lagemann, C., Fasel, U., Kutz, J. N., & Brunton, S. L. (2025). SINDy-RL for interpretable and efficient model-based reinforcement learning. *Nature Communications*, 16(1), 10714.
- Mason, J. J., Allen-Blanchette, C., **Zolman, N.**, Davison, E., & Leonard, N. E. (2023). Learning to predict 3d rotational dynamics from images of a rigid body with unknown mass distribution. *Aerospace*, 10(11), 921.
- Marcolli, M., & **Zolman, N.** (2019). Adinkras, dessins, origami, and supersymmetry spectral triples. *p-Adic Numbers, Ultrametric Analysis and Applications*, 11(3), 223–247.
- Ma, X., Hopkins, P. F., Faucher-Giguère, C.-A., **Zolman, N.**, Muratov, A. L., Kereš, D., & Quataert, E. (2016). The origin and evolution of the galaxy mass–metallicity relation. *Monthly Notices of the Royal Astronomical Society*, 456(2), 2140–2156.

### PREPRINTS

- Howard, A. A., **Zolman, N.**, Jacob, B., Brunton, S. L., & Stinis, P. (2026). SINDy-KANs: Sparse identification of non-linear dynamics through Kolmogorov-Arnold networks. *arXiv preprint arXiv:2603.18548*.
- Zolman, N.**, Mokbel, S., Otto, S. E., & Brunton, S. L. (2026). Uncovering extreme event mechanisms for prediction and control with sensitivity-balanced projections. *arXiv preprint arXiv:2606.05618*.
- Andrew, K., Steinfelds, E., & **Zolman, N.** (2016). The Mattig expression for a d-dimensional Gauss Bonnet FRWL cosmology. *arXiv preprint arXiv:1601.07037*.

## Awards, Fellowships, & Grants

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- 2026    **BARC Research Excellence Team Award**, Boeing Advanced Research Center
- Awarded to the Data-Driven Engineering Enabling R&D Team for advancing physics-informed, data-driven engineering research through the UW-Boeing partnership.
- 2022    **Aerospace Graduate Fellowship Program**, The Aerospace Corporation
- Up to \$15k for tuition towards graduate studies per year. Renewal up to 5 years.
- 2016    **Mari P. Ligocki Award**, California Institute of Technology
- “Given to a student who has improved the quality of student life at Caltech through their personal character.”
  - Awarded by the Deans Office for contributions to the Upperclassman Counselor (UCC) program, role on the Title IX Advisory Board, and annual involvement in freshman orientation.

## Presentations

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### INVITED TALKS

- April 2026: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk (virtual). The Boeing Company Physics-Informed Machine Learning Community of Interest. Everett, WA, USA.

September 2025: *Symmetry Enforcement, Discovery, and Promotion for ML Applications in Engineering*. Invited talk. Harvard University Geometric Machine Learning Group. Cambridge, MA, USA.

September 2025: *Symmetry Enforcement, Discovery, and Promotion for ML Applications in Engineering*. Invited talk. MIT Atomic Architects Group. Cambridge, MA, USA.

March 2025: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk (virtual). MIT Mobility Lab. Cambridge, MA, USA.

October 2024: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk (virtual). University of Sydney Brain and Mind Centre. Sydney, Australia.

February 2024: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk. Imperial College London. Department of Aeronautics. London, United Kingdom.

February 2024. *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk. University of Cambridge Applied and Computational Analysis series. Cambridge, United Kingdom.

## CONTRIBUTED PRESENTATIONS

July 2026: *Characterizing Extreme Events in Dynamical Systems through Sensitivity-Based Modal Decomposition*. Oral Presentation. WCCM/ECCOMAS 2026. Munich, Germany.

March 2026: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation. ERCOFTAC ML4Fluids Workshop. Amsterdam, Netherlands.

November 2025: *Characterizing Extreme Events in Turbulent Flows through Sensitivity-Based Modal Decomposition*. Oral Presentation. APS DFD. Houston, TX, USA.

November 2025: *Characterizing Extreme Events in Turbulent Flows through Sensitivity-Based Modal Decomposition*. Oral Presentation. UW Computational Neuroscience Center Minisymposium on State Space Models and Dynamical Systems. Seattle, WA, USA.

May 2025: *Symmetry Enforcement, Discovery, and Promotion for Machine Learning Applications in Engineering*. Oral Presentation. SIAM Applications of Dynamical Systems Minisymposium: Data-Driven Modeling and Estimation for Complex Systems. Denver, CO, USA.

March 2025: *Data-Driven Reduced Order Modeling of Extreme Events in Turbulent Flows*. Oral Presentation. CFC Minisymposium: Physics-based and Data-driven Low-order Modeling for Turbulent Flows. Santiago, Chile.

February 2025: *Data-Driven Dimension Reduction Through Symmetry-Promoting Regularization*. Oral Presentation. DTE-AICOMAS Minisymposium: Deep Learning-based Reduced Order Models in Scientific Computing. Paris, France.

January 2025: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation and Panelist. Joint Mathematics Meeting Minisymposium: Scientific Machine Learning: Recent Advances and Future Directions. Seattle, WA, USA.

November 2024: *Data-Driven Dimension Reduction Through Symmetry-Promoting Regularization*. Oral Presentation. APS DFD. Salt Lake City, UT, USA.

July 2024: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation. WCCM Minisymposium: Synergistic computational mechanics + machine learning for the digital twinning of intelligent vehicles. Vancouver, BC, Canada.

May 2024: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation. ECCOMAS Minisymposium: Recent Advances in Deep RL of Complex Dynamical Systems. Lisbon, Portugal.

February 2024: *SINDy-RL: Uncertainty-aware active learning for simultaneous dynamics discovery and control*. Oral Presentation. SIAM-UQ Minisymposium: Reduced order modeling, Learning, UQ, and their interaction. Trieste, Italy.

November 2023: *SINDy-RL: Interpretable and efficient reinforcement learning for fluid flow control*. Oral Presentation. APS DFD. Washington D.C., USA.

June 2023: *Reinforcement Learning Tutorial*. Oral Presentation. NSF AI Institute in Dynamic Systems: Machine Learning Workshop. Seattle, WA, USA.

July 2019: *Application of reinforcement learning to rendezvous and proximity operations*. Oral Presentation and Panelist. Seventh International Conference on Space Mission Challenges for Information Technology Mini-Workshop: Machine Learning for Spaced-Based Applications. Pasadena, CA, USA.

## Teaching Experience

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|-----------|-------------------------------------------------------------------------|----------------------------|
| 2026 (SP) | <b>AE 598: Intro to AI for Aerospace Engineers</b> , Guest Lecturer     | <i>Univ. of Washington</i> |
| 2026 (SP) | <b>ME 569: Data-Driven Control</b> , Guest Lecturer                     | <i>Univ. of Washington</i> |
| 2025 (SU) | <b>Youth Camp: Introduction to Math Modeling</b> , Guest Lecturer       | <i>Univ. of Washington</i> |
| 2025 (SP) | <b>ENGR 521: Physics-Informed Machine Learning</b> , Guest Lecturer     | <i>Univ. of Washington</i> |
| 2018 (SP) | <b>Math 134: Introduction to Dynamical Systems</b> , Teaching Assistant | <i>UCLA</i>                |
| 2018 (WI) | <b>Math 132: Complex Analysis for Applications</b> , Teaching Assistant | <i>UCLA</i>                |
| 2017 (AU) | <b>Math 33B: Differential Equations</b> , Teaching Assistant            | <i>UCLA</i>                |

## Mentoring

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| 2026–     | <b>Krithik Senthilkumar</b> , High School Student, University of Washington<br>• Guided Reading: Active learning for SINDy and control.                 | <i>Seattle, WA</i>    |
| 2025–     | <b>Anuj Krovvidi</b> , Undergraduate Research Assistant, University of Washington<br>• Project: Direct policy optimization with SINDy surrogate models. | <i>Seattle, WA</i>    |
| 2022      | <b>Eric Ma</b> , Undergraduate Intern, The Aerospace Corporation<br>• Project: Dynamics extraction from neuromorphic sensor data.                       | <i>El Segundo, CA</i> |
| 2021      | <b>Noah Haig</b> , Undergraduate Intern, The Aerospace Corporation<br>• Project: Physics-informed neural networks for anomaly detection.                | <i>El Segundo, CA</i> |
| 2021      | <b>Austin Marstaller</b> , Graduate Intern, The Aerospace Corporation<br>• Project: Physics-informed neural networks for anomaly detection.             | <i>El Segundo, CA</i> |
| 2020–2022 | <b>Justice Mason</b> , Graduate Intern, The Aerospace Corporation<br>• Project: Learning to predict 3D rotational dynamics from images of a rigid body. | <i>El Segundo, CA</i> |
| 2020      | <b>Jalani Williams</b> , Graduate Intern, The Aerospace Corporation<br>• Project: Hamiltonian neural networks for rotating bodies.                      | <i>El Segundo, CA</i> |
| 2020      | <b>Gaurav Kuppa</b> , Undergraduate Intern, The Aerospace Corporation<br>• Project: Deep reinforcement learning for robotic proximity operations.       | <i>El Segundo, CA</i> |
| 2020      | <b>Dhruv Patel</b> , Graduate Intern, The Aerospace Corporation<br>• Project: Topological data analysis for time series anomaly detection.              | <i>El Segundo, CA</i> |

## Outreach & Professional Development

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### SERVICE AND OUTREACH

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| 2022–     | <b>UW Women of Aerospace</b> , Mentor<br>• 2024-25 Mentee: Neha Yelve<br>• 2022-24 Mentee: Jerusalem Sintayehu |
| 2020–2022 | <b>Aerospace Corporation Physics + ML Journal Club</b> , Founder and coordinator                               |
| 2019–2022 | <b>Aerospace Corporation Intern Program</b> , Mentor, panelist, and coordinator.                               |
| 2014–2016 | <b>Caltech Title IX Advisory Board</b> , House Lead                                                            |
| 2013–2015 | <b>Caltech Upperclassman Counselor</b> , Head UCC                                                              |

### ORGANIZATION AND LEADERSHIP

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| 2026 | <b>WCCM-ECCOMAS 2026</b> , Minisymposium Organizer and Session Chair<br>• MS: <i>Cents &amp; Sensitivities: Exploiting Differentiable Solvers for Low-Cost Optimization in Science and Engineering</i> | <i>Munich, Germany</i> |
| 2025 | <b>DTE-AICOMAS 2025</b> , Session Chair<br>• MS: <i>Advancements of Data-Driven Methods in Computational Mechanics</i>                                                                                 | <i>Paris, France</i>   |

- 2025– **NSF AI Institute in Dynamic Systems**, Seminar organizer and host *Seattle, WA*
- Hybrid Seminar Series: *Data-Driven Science and Engineering*
  - Identified and invited speakers from a variety of academic backgrounds
  - Hosted weekly seminars, moderated in-person and virtual questions.
  - Organized in-person research meetings, lunch with students, and dinner with faculty.
- 2023– **Data-Driven PhD “Stand-up”**, Founder and organizer *Seattle, WA*
- Led weekly meeting for graduate students across different groups focused on using ML in engineering.
  - Meetings focused setting intentional goals, troubleshooting problems, and fostering community.

## PEER REVIEW

Proceedings of the National Academy of Sciences  
Physical Review Letters Fluids  
Learning for Dynamics and Control Conference (L4DC)

## PROFESSIONAL MEMBERSHIPS

Society of Industrial and Applied Mathematicians  
American Physical Society